TraitHorizon: Scalable Exploration of Large Image-Feature Paired Datasets

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1. Summary

Collections of Objects of Interest (OOIs)—such as cells, tubules, or tissue patches—paired with high-dimensional, computationally derived feature vectors (e.g., morphology, texture, or deep-learned embeddings) are increasingly being generated in image-based biomedical research. These OOI-feature datasets are routinely explored to detect outliers for quality control, uncover patterns, and generate biological insights (e.g., identifying distinct morphological patterns among tubules). However, as these datasets grow in size and complexity, researchers face several bottlenecks: repeated manual creation of static visualizations for distinct OOI subsets (e.g., normal vs atrophic tubules), difficulties in efficiently plotting large numbers of high-dimensional feature vectors, and limited tooling for tracing individual feature values back to their source objects. To address these challenges, we developed TraitHorizon, a browser-based visualization platform for interactive exploration of large OOI–feature pair datasets. TraitHorizon integrates three synchronized visualization components: (a) a parallel coordinates plot for feature-vector–level exploration, (b) dynamic violin plots for per-feature distribution analysis, and (c) a tabular data grid linking each feature vector to its corresponding object. Its interactive interface enables real-time, scalable visualization of datasets containing hundreds of thousands of OOI-feature pairs. We demonstrate TraitHorizon's utility through a case study involving quality control of a digital pathology dataset comprising 260,201 segmented tubules in kidney biopsies, each characterized by 99 features, illustrating how the platform facilitates rapid, interpretable, and reproducible data interrogation.

TraitHorizon is publicly available for use and modification at https://traithorizon.com/.

2. Statement of Need

Research in image-based fields (e.g., computational pathology) often aims to discover trends in Objects of Interest (OOIs), associated with diagnosis, prognosis, and therapy response. Objects can consist of individual items (e.g., cells), higher-order structures (e.g., tubules or infiltration patterns), or even small image patches. From these OOIs, sets of numerical features termed feature vectors are often extracted

to encode their attributes. Features can range from simple (e.g., area, stain intensity, texture) to more complex, hand-crafted (e.g., aspect ratio of peritubular capillaries (Yijiang Chen et al. 2025)), or even deep learning-derived descriptors (Yijiang Chen et al. 2025; Fan Fan et al. 2025; Y. Chen et al. 2023; Echle et al. 2021; Ambekar et al. 2025). These features can then be examined to assess their associations with object phenotypes—such as biological signals (e.g., disease state) or image quality factors (e.g., blurriness)—or how they positively/negatively correlate with one another.

Such analyses often involve generating static, non-interactive visualizations, such as parallel coordinates plots (Heinrich and Weiskopf 2013) or scatter plots with UMAP (McInnes, Healy, and Melville 2020) or t-SNE (Maaten and Hinton 2008). Aided by first-order summary statistics (e.g., mean, standard deviation), these visualizations are inspected to uncover structure and outliers within the feature distribution. Multiple data-driven subpopulations often emerge from this initial exploration, prompting repeated generation of visualizations and summary metrics for each subpopulation. Quantitative and qualitative patterns uncovered within subpopulations can, in turn, guide comparative analyses—for example, assessing how a given feature or pattern differs between "diseased" and "normal" groups.

However, to date, this investigatory workflow remains time-intensive due to three main limiting factors:

1. **Lack of connection between an object's image, feature values, and cohort-level visualization.** Analysts often discover outliers or clusters within cohort-level visualizations, prompting the need to trace them back to the associated images for visual inspection. Static plots can at best display only a modest number of images, forcing analysts to manually map plotted points to objects, incurring significant time costs. The ability to instantly view the object associated with a plotted feature vector is essential for rapid validation, interpretation, and insight generation.
2. **Exploring each subpopulation incurs high time cost.** Biomedical imaging datasets often include heterogeneous subpopulations—such as tissue types, disease subtypes, or experimental conditions—that warrant the generation of separate visualizations for detailed inspection. Manually plotting each subpopulation becomes inefficient in repetitive workflows, while programmatic (e.g., loop-based) plot generation requires subpopulations of interest to be identified and well-defined *a priori*. Dynamic filtering and responsive plots are necessary for users to iteratively explore subpopulations without incurring substantial time costs from repeated plot generation and review. Furthermore, the conditions/filters used to identify them should be easy to document, save, and reapply.
3. **Latency while rendering large datasets disrupts analysis.** OOI-feature datasets can easily reach hundreds of thousands of data points. When subpopulations are not yet clearly defined, a "global" view of the entire dataset is required before smaller subpopulations can be explored. Browser-based tools not designed for plotting at this scale can suffer from heavy memory utilization, latency, and even unresponsiveness/browser crashes while rendering such data visualizations. A capable visualization tool must leverage rendering techniques that remain efficient and within browser limitations as the number of objects scales.

These limiting factors motivate the development of interactive data visualization tools that can dynamically generate visualizations as the user explores their data. However, existing tools do not fully address all three limiting factors simultaneously. TensorBoard Projector (TensorFlow Team 2025) and HoloViews ("Installation — HoloViews V1.21.0" n.d.) support interactive plotting of large multivariate datasets but do not provide for image viewing. Conversely, DendroMap (Bertucci et al. 2022) facilitates qualitative exploration of large sets of images but does not link back to their feature vectors. These tools also do not natively support dynamic filtering by feature value. HistoQC (Janowczyk et al. 2019) provides image-linked plots and metrics, but it is tailored specifically to quality control use cases in digital pathology and thus does not support arbitrary user-defined feature vectors or images.

To address these gaps, we introduce TraitHorizon, a browser-based application for interactive exploration of images alongside their feature vectors. TraitHorizon (a) directly links plotted data points to their source object images, enabling rapid validation and interpretation, (b) supports dynamic, multi-dimensional filtering with support for saving and loading filter configurations, and (c) efficiently renders hundreds of thousands of data points in-browser. By doing so, TraitHorizon facilitates efficient exploratory analysis in object-based research workflows.

3. Implementation

TraitHorizon is a Flask-based application ("Pallets/Flask: The Python Micro Framework for Building Web Applications." n.d.), making it ideal for collaborative environments when hosted over a network. The front end is built with JavaScript, HTML5, and CSS, and leverages visualization libraries such as SlickGrid ("SlickGrid Home" n.d.), Parallel Coordinates (parcoords (Yun 2025)), and D3.js ("D3 by Observable | The JavaScript Library for Bespoke Data Visualization" n.d.) to power its interactive dashboard. Its command-line interface ingests a single directory of object image files (e.g., .png, .jpg, .svg, or any other browser-supported format ("Image File Type and Format Guide - Media | MDN" 2025)) and a tab-separated values (TSV) file, with each row containing an image file path followed by a user-determined number of tab-separated feature columns. As such, any tool outputting tabular data is compatible with TraitHorizon's simple input format (e.g., HistoQC (Janowczyk et al. 2019) and CohortFinder (Fan Fan et al. 2024)). TraitHorizon supports integer, float, scientific notation, and categorical features. Clickable URLs linking to external web-based tools are supported by an optional column titled "url".

TraitHorizon is designed to meet the demands of datasets containing hundreds of thousands of objects. To minimize browser lag during plotting, five application features were implemented:

1. Non-blocking progressive rendering occurs in the parallel coordinates plot, completing in seconds to minutes depending on dataset size. Importantly, the user interface remains responsive during this rendering period, allowing users to continue interacting with the parallel coordinates plot and other functional components without straining browser responsiveness.
2. Parallel coordinate filter configurations and associated filtered object IDs can be exported and imported in a lean JSON format, without requiring a re-render of the parallel coordinates plot.
3. The data grid supports both pagination and alternatively "infinite scrolling". This web design technique enables new content to be dynamically loaded, replacing old content as the user scrolls down and creating the illusion of a never-ending page within a fixed memory footprint.
4. Object images are loaded on-demand to limit browser memory, network utilization, and CPU usage.
5. Violin plots are also computed on-demand when the user hovers over a parallel axis, avoiding CPU usage associated with refreshing violin plots each time the parallel coordinates plot is updated.

These design choices allow TraitHorizon to operate smoothly within the constraints of modern web browsers and modest consumer-grade hardware, regardless of dataset size.

4. Use Case – Interactive Exploration of Tubular Pathomic Features via TraitHorizon

To illustrate TraitHorizon's functionality, we provide the following detailed use case from a recent digital pathology image-based biomarker study (Fan Fan et al. 2025; F. Fan et al. 2024). In our use case, we focused on data exploration of 260,201 segmented tubules (Fogo et al. 2016) and their associated features (for a total of 99 features per tubule) as computed from 254 Periodic Acid-Schiff (PAS) stained kidney biopsies from the Nephrotic Syndrome Study Network (NEPTUNE) (Barisoni et al. 2013; Gadegbeku et al. 2013) and Cure Glomerulonephropathy (CureGN) ("CureGN Study Rationale, Design, and Methods: Establishing a Large Prospective Observational Study of Glomerular Disease - PubMed" n.d.) consortia. The goal was to support quality control and uncover tubular morphological patterns along a pathway from normal to tubular atrophy (Fogo et al. 2016; "Automated Computational Detection of Interstitial Fibrosis, Tubular Atrophy, and Glomerulosclerosis | American Society of Nephrology" n.d.).

4.1. Overview and Dataset Preparation

Before running TraitHorizon, users must prepare a TSV file containing the following columns:

1. **"filename" (required)** – Specifies the base filename for each instance image, without the folder path (e.g., image1.png).

2. **Features (required)** – A tab separated row containing all features associated with each instance. Our dataset included 99 tubular pathomic features capturing morphological and topological characteristics of the basement membrane, epithelium, nuclei and lumen (Fan Fan et al. 2025). In our pre-TraitHorizon analysis, non-negative matrix factorization (Lee and Seung 2001; F. Fan et al. 2025) (rank = 14) followed by UMAP on these features revealed 1,667 tubules forming outlier clusters. These were flagged in a separate "Needs QC" column.
3. **"url" (optional)** – Allows linking each object to external resources such as secondary visualizations, related analyses, or web-based viewers. In our use case, these URLs open the corresponding tubule in the HistomicsUI viewer ("DigitalSlideArchive/HistomicsUI" 2025) within the Digital Slide Archive (DSA) (Gutman et al. 2017), which displays the tubule's location within the original tissue slide for added histologic context.

4.2. Running TraitHorizon

The folder containing all image files should be provided to TraitHorizon's command-line interface via the `assets_path` argument. All images must reside in this directory, which TraitHorizon uses to locate and display them in the user interface. Similarly, the path to the TSV file should be specified using the `tsv_path` argument.

After starting the TraitHorizon server, a local URL (e.g., http://localhost:5000) will appear in the terminal. Opening this URL in a web browser launches the TraitHorizon user interface (Figure 1), allowing users to interactively explore images and their associated features.

4.3. Initial Exploration with Parallel Coordinates Plots and Data Grid

Initial exploration began with the parallel coordinates plot, which provided a global view of feature distributions across hundreds of thousands of tubules (Figure 1-A). By plotting the N-dimensional feature vector of each OOI as a continuous line intersecting N respective axes, the parallel coordinates plot preserves both the full dimensionality of feature vectors (unlike dimensionally-reduced scatter plots) and the relationships between intra-OOI feature values (unlike marginal views such as violin plots). When hovering over a feature axis, a violin plot is displayed (Figure 1-B), enabling distribution visualization and noting of properties such as skewness and normality. Here, when hovering over the feature `TBM_THICK_MAX` (maximum tubular basement membrane thickness), the violin plot revealed a strongly skewed distribution. This suggested the presence of tubules with extremal values, which required closer examination in the data grid (Figure 1-C) for qualitative assessment. Together, these visualizations help users quickly identify whether features are broadly consistent, contain outliers, or show irregular patterns that may reflect artifacts or biologically distinct characteristics.

4.4. Filtering Based on Feature(s)

TraitHorizon's interactive filtering feature allows users to isolate and inspect data subpopulations based on feature values. For example, we filtered tubules with high `TBM_THICK_MAX` values (maximum tubular basement membrane thickness) by dragging a brush (Yun 2025) along the corresponding axis in the parallel coordinate plot (Figure 2 – green box). The filtering action highlighted the relevant instances (tubules) in red and displayed their rows in the data grid, where we observed that these tubules consistently showed thickened basement membranes (large green segments), a hallmark of atrophic tubules.

TraitHorizon's data grid also displays images alongside their respective quantitative features, allowing users to explore individual instances in greater detail without switching contexts to perform image lookups. Using the data grid we visually examined the subset of tubules with high `TBM_THICK_MAX` values and found an unexpected phenotype characterized by enlarged lumina. When hovering over these rows in the data grid, the corresponding feature vectors were highlighted in blue within the parallel coordinates plot (Figure 3-A). Comparing the qualitative images with corresponding feature vectors suggested the existence of a small subpopulation of tubules exhibiting both thickened TBM and lumen dilation (moderately elevated values of `LUMEN_THICK_MAX`, the maximum lumen thickness; Figure 3-B). To further study this subpopulation, we

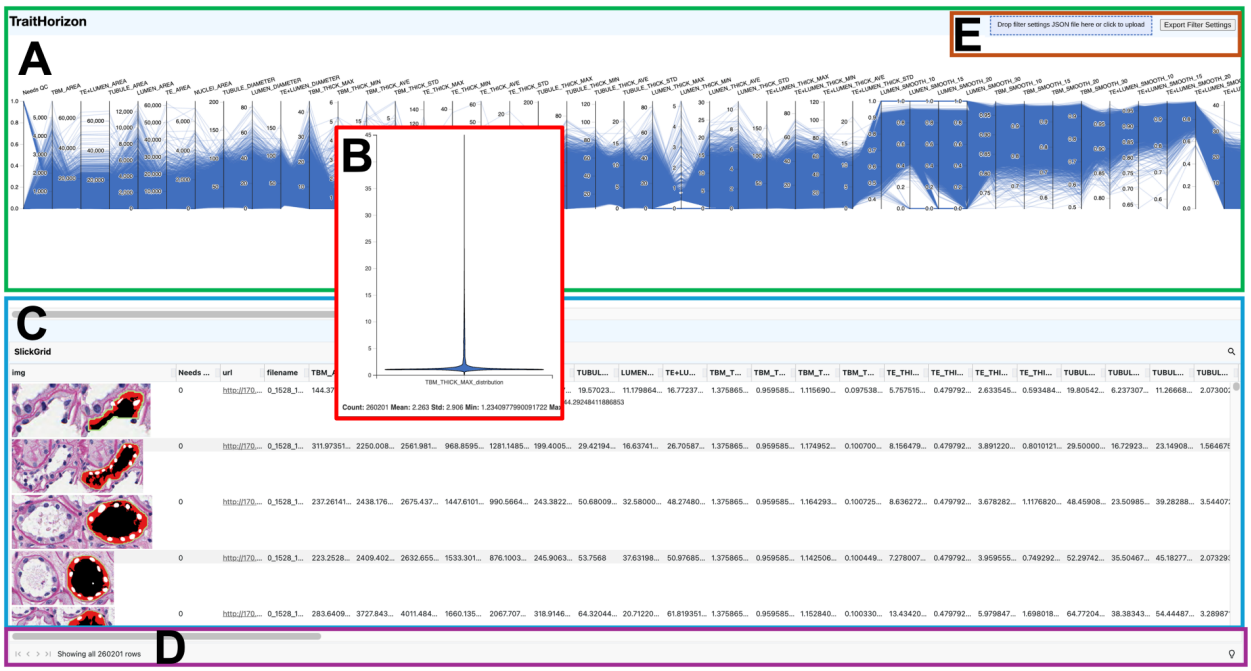


Figure 1: TraitHorizon web application interface. (A) The interactive parallel coordinates plot displays 99 pathomic features from 260,201 tubules, wherein each blue line represents one tubule feature vector. (B) A violin plot shows the value distribution of a selected feature when hovered over. (C) The data grid presents each segmented tubular image, with the original PAS-stained tubule on the left and the segmentation overlay on the right (green: tubular basement membrane; white: tubular nuclei; black: tubular lumen; red: tubular epithelium). Associated feature data loaded from the TSV (e.g., filename and corresponding feature values) are also displayed. (D) The bottom status bar shows the number of instances currently visible. (E) The drop zone and export button allow users to save filter configurations and filtered row IDs. TraitHorizon allows users to customize which columns are included or excluded in the parallel coordinates plot. Here, the URL and filename columns were excluded from the parallel coordinates plot but are shown in the data grid.

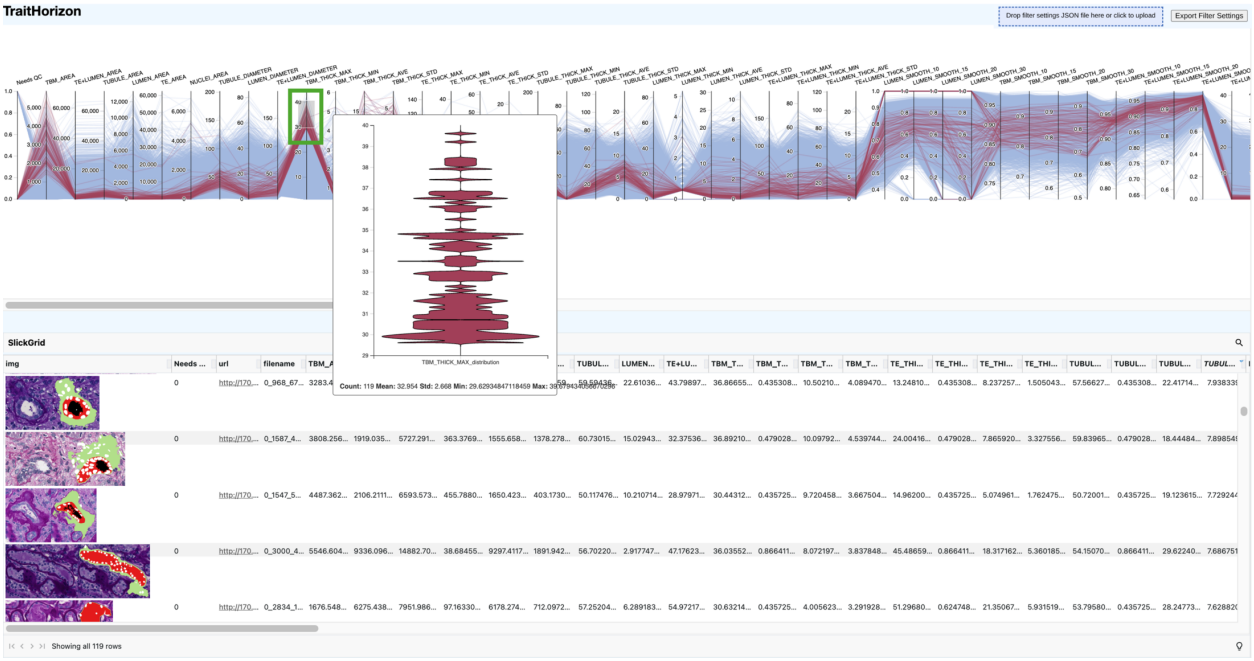


Figure 2: Interactive filtering with only one feature. TraitHorizon allows users to interactively filter instances based on any feature shown in the parallel coordinates plot. In this example, we hovered over the axis for the feature `TBM_THICK_MAX` (maximum basement membrane thickness) and dragged a brush to isolate tubules with values in the range from 30 to 40. In the parallel coordinate plot, 119 instances were dynamically highlighted in red after filtering (with this number indicated in the status bar), while all other ones remained visible but are greyed out. This color contrast makes it easier to visually compare the selected subset with the rest of the dataset across all other features. As shown here, the selected tubules were associated with visibly thickened basement membranes (enlarged green segments). Such interactive visualization enables users to explore and validate morphological patterns interactively. When hovering over a feature axis, a violin plot is generated for the selected subset (in red), providing an overview of feature-value distributions. In the data grid, only the filtered instances are shown.

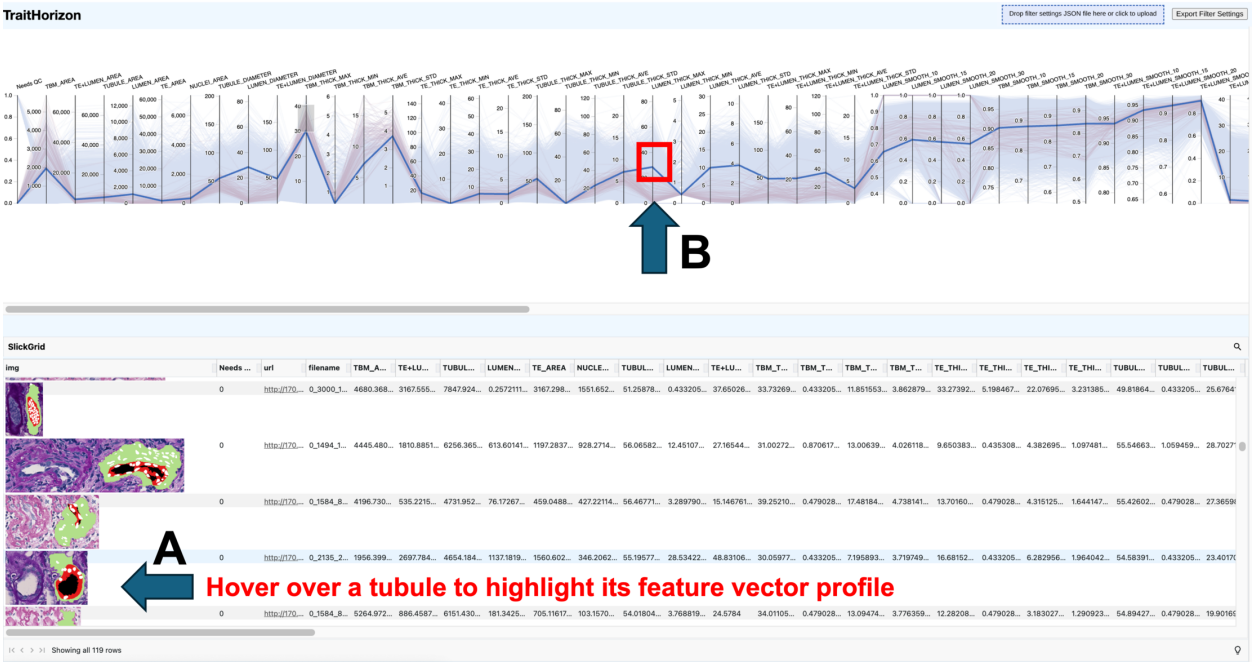


Figure 3: TraitHorizon enables the identification of multi-feature expression patterns. To inspect a specific instance, users can first (A) hover over the instance in the data grid, which highlights the corresponding row in both the parallel coordinates plot and the data grid in blue. By visually examining its feature vector, users can identify features of interest. In this example, the selected tubule showed high expression of `TBM_THICK_MAX`, and (B) moderate expression of `LUMEN_THICK_MAX` (maximum thickness of the tubular lumen).

applied an additional brush on `LUMEN_THICK_MAX` in the parallel coordinates plot (Figure 4-2). Subsequent review with study pathologists indicated that these tubules represent a composite morphological phenotype, combining basement membrane thickening with an enlarged lumen (Fogo et al. 2016).

In addition, TraitHorizon allows users to export and import filters for reproducibility and record-keeping. Specifically, clicking the "Export Filter Settings" button results in the filtered row IDs and corresponding filter values being saved as a JSON file (Figure 4-3). Dragging the JSON file into the drop zone (Figure 4-4) results in the brush being automatically restored and the corresponding subset of objects being reshown in the data grid.

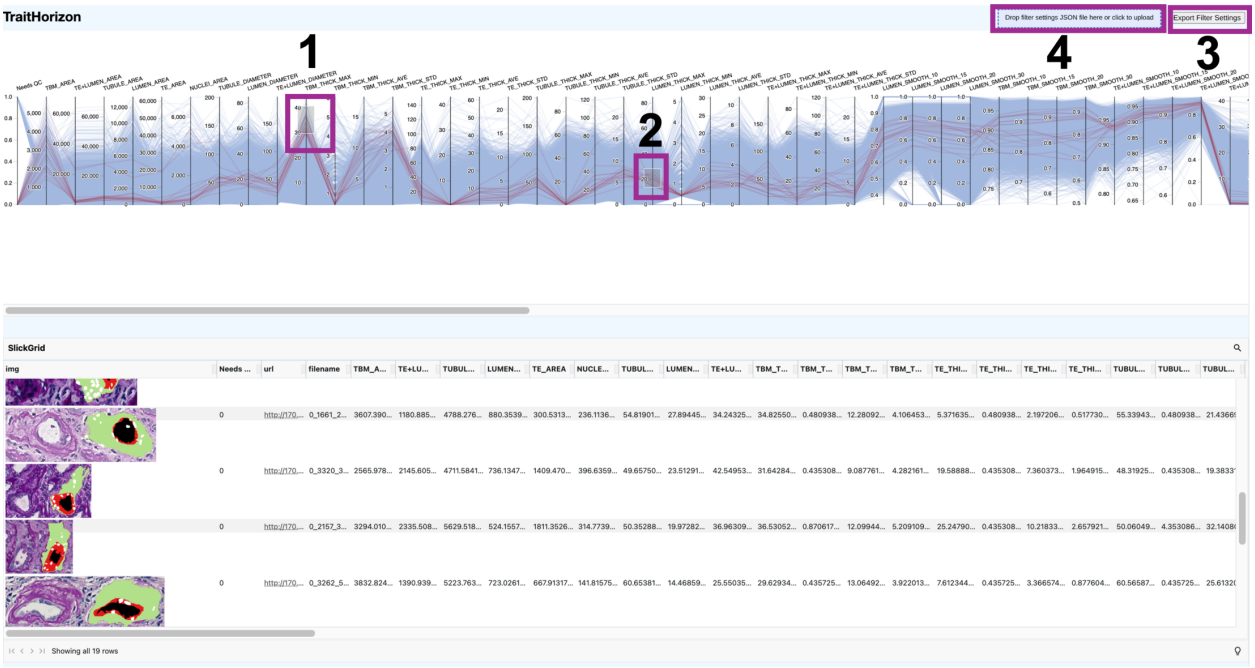


Figure 4: Interactive joint filtering with multiple features. TraitHorizon supports joint filtering across multiple features to identify more complex patterns. In this example, we first (1) dragged a brush to isolate tubules with `TBM_THICK_MAX` values between 30 and 40, and then (2) applied a second brush on `LUMEN_THICK_MAX`, selecting values between approximately 15 and 25. This combined filtering enabled the identification of tubules showing both basement membrane thickening and enlarged lumen. Users can export filter configurations and filtered row IDs as a JSON file by clicking the (3) 'Export Filter Settings' button. For reproducibility, the exported JSON can later be reloaded by dragging it into the (4) drop zone, automatically restoring the filtering criteria.

Overall, the seamless integration of feature visualization, interactive filtering, and image review allows users to (1) rapidly assess whether unusual feature values reflect histopathological patterns or segmentation/technical artifacts, (2) uncover novel biological insights (e.g., complex disease phenotypes) that might be missed using univariate methods, ultimately supporting more informed QC and pattern exploratory analysis, (3) enhance reproducibility by enabling the storage, sharing, and restoration of filtering configurations for downstream analyses.

4.5. External URL Integration for Contextual Visualization in HistomicsUI

TraitHorizon supports the integration of external URLs for each instance, enabling users to link directly to additional resources such as image viewers, annotation tools or clinical databases, depending on the needs of the workflow (Figure 5). In our implementation, this feature was used to connect TraitHorizon with DSA. Clicking the URL associated with each tubule (Figure 5-1) opened the corresponding region in the DSA's viewer (named HistomicsUI) at the WSI level (Figure 5-2), with the tubule centered in view. This allowed

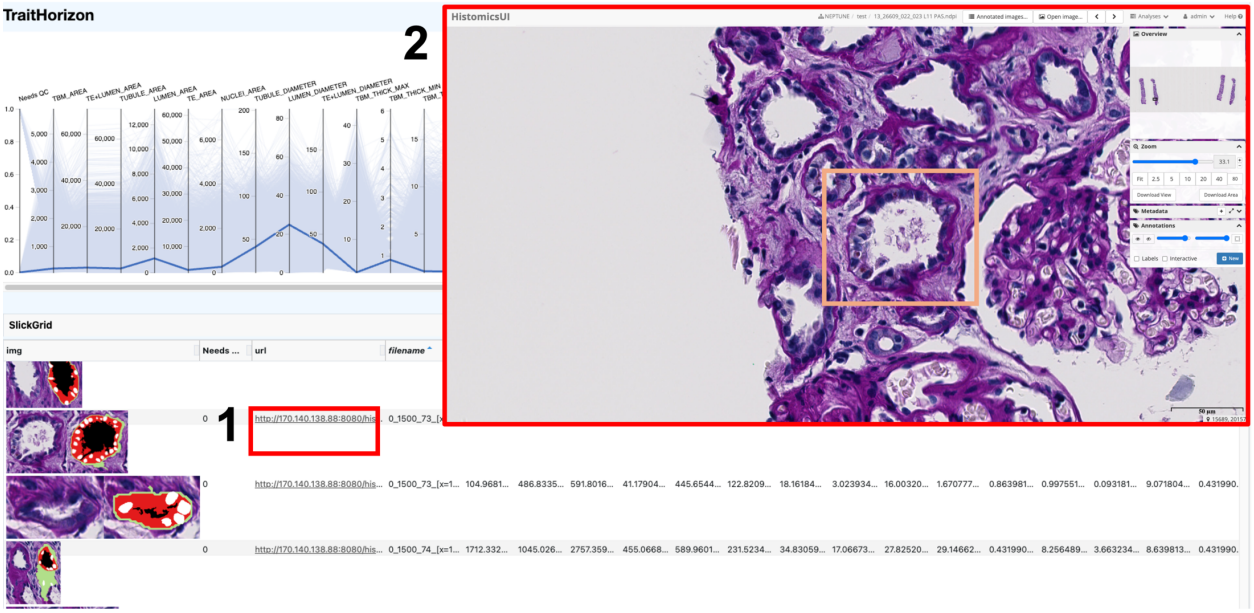


Figure 5: TraitHorizon supports integration with external viewers such as HistomicsUI through user-defined URL links. In this example, the link to the DSA has been pre-configured so that clicking the "url" cell of the row (1) opened an external browser window displaying the target tubule centered in the DSA's HistomicsUI viewer (2). This functionality can also be used to connect objects to other external resources, such as clinical data, supplementary analyses, or metadata information.

pathologists to assess the broader tubule microenvironment—including adjacent tubules, interstitial areas, and glomeruli—rather than solely examining isolated objects. Notably, this feature is not limited to external image viewers but can be extended to any external resource to support different analytical workflows.

4.6. Searching by Exact Value

Beyond numerical range filtering, TraitHorizon also supports searching by specifying exact values for selected features. Users can (1) click the magnifier icon (Figure 6-1), (2) select the desired feature, and (3) input a specific category or value to search the dataset (Figure 6-2). Using this functionality, we searched for the "Needs QC" indicator to investigate tubules flagged for review. Visual inspection revealed that many flagged tubules lacked a visible lumen, and quantitative analyses (via the parallel coordinates plot) confirmed consistently low values for lumen-related features such as `LUMEN_THICK_AVE` (average lumen thickness, Figure 6-3). TraitHorizon thus enables rapid identification and efficient validation of instances with specific feature values—something that is often cumbersome in workflows using traditional/static tools.

Importantly, this search capability is not restricted to quality control indicators. It can be applied to any features in the dataset, enabling users to flexibly isolate and analyze subsets of interest based on labels, classifications, or other features. This is particularly useful for features containing numerous unique categories or identifiers—for example, identifying a specific tubule by its filename or retrieving all tubules originating from the same patient.

4.7. Sorting Numerical Features

TraitHorizon also supports sorting of numerical features directly within the data grid by clicking on a column header (Figure 7). In our workflow, we sorted tubules by `TBM_AREA` (tubular basement membrane area) to explore morphological variation. Scrolling through the sorted grid revealed a clear gradient in morphology—specifically, progressive thickening of the basement membrane—enabling intuitive observation of trends from normal to atrophic tubules.

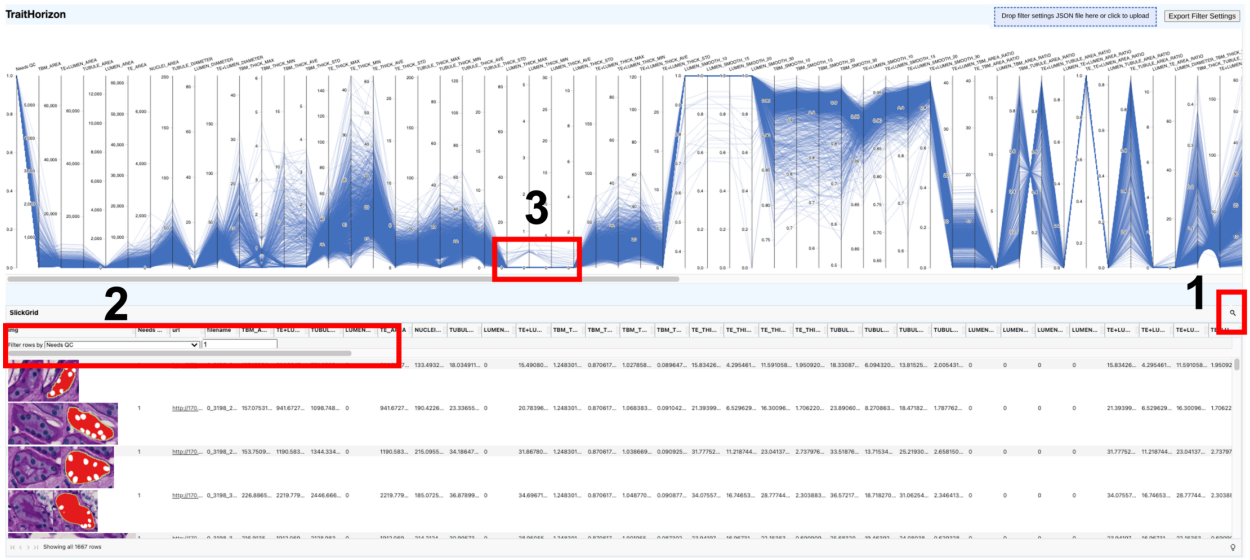


Figure 6: Flexible feature search. Using the user-defined "Needs QC" indicator (where '1' marks tubules flagged for quality control), we (1) clicked the magnifier icon, and (2) manually entered the flag value to search the dataset. (3) The subset highlighted tubules lacking lumens, as evidenced by low values across lumen-related features such as `LUMEN_THICK_AVE/MIN/MAX/STD` (average, minimum, maximum, and standard deviation of tubular lumen thickness). This search functionality enables rapid identification of instances with specific feature values that need further review.



Figure 7: Sorting numerical features. TraitHorizon allows users to sort numerical features directly within the data grid by clicking on a column header. Features can be sorted in either descending (left) or ascending (right) order. In this example, the feature `TBM_AREA` (tubular basement membrane area) was sorted to facilitate comparison between tubules with large/thickened basement membranes and those with small/thin ones, helping users visually assess underlying morphological differences. In the descending view (left panel), the top-ranked tubules exhibited larger or thickened TBM regions (extensive green segments). Conversely, in ascending view (right panel), the top entries corresponded to tubules with smaller or thinner basement membranes. By sorting features interactively, users can identify qualitative gradients across ordered set of objects, revealing progressive feature changes within the dataset.

5. Discussion and Conclusions

TraitHorizon addresses a critical gap in image-based data analysis by enabling fast, interactive, and image-linked exploration of high dimensional feature datasets. Traditional visualization approaches often fail to couple image data with derived features and support interactive plotting at scale. In contrast, TraitHorizon links each source image to its high dimensional feature vector, supports real-time interactive filtering and sorting, and utilizes efficient rendering strategies to scale to datasets with hundreds of thousands of images.

TraitHorizon's modular implementation is well suited for future development to improve existing limitations. For instance, (a) making the parallel coordinates plot resizable and adding style customization options would allow for user adjustments to alleviate overplotting, and (b) adding an enlarged image view would make image inspection more convenient.

TraitHorizon's features empower researchers to discover meaningful patterns, perform quality control, and generate insights for large object-feature datasets. Furthermore, TraitHorizon's support for arbitrary features, multiple feature types, and external URLs enables its broad use across multiple computational imaging fields.

TraitHorizon is available as a python package, and a Docker image is released for easy installation (https://traithorizon.com/).

6. AI usage disclosure

No generative AI tools were used in the development of the TraitHorizon software or for writing the original manuscript. Claude Opus 4 (Anthropic) was used solely to assist with restructuring and condensing the manuscript to meet the updated JOSS format and word limit requirements. Github Copilot was used to convert the TraitHorizon documentation from .rst format to markdown. All AI-generated text was reviewed, edited, and validated by the authors, who made all decisions regarding scientific content and framing.

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